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Protective barrier for safety glazing

Abstract

A protective barrier affixable to a curved substrate comprises a stack of two or more lenses, each of the two or more lenses including a polyethylene terephthalate (PET) film, a hard coat on a first side of the PET film, and an adhesive layer on a second side of the PET film opposite the first side. The stack of two or more lenses may have a modulation transfer function that exhibits a contrast value greater than 75% for a spatial resolution of one line-pair per 0.0003 radians at 65 degrees angle of incidence. Heat and pressure may be applied to conform the stack of two or more lenses to the shape of the curved substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/786,149, filed Jul. 26, 2024, which is a continuation of U.S. application Ser. No. 18/478,220, filed Sep. 29, 2023, now U.S. Pat. No. 12,077,037, issued Sep. 3, 2024, which is a continuation of U.S. application Ser. No. 18/062,901, filed Dec. 7, 2022, now U.S. Pat. No. 11,807,078, issued Nov. 7, 2023, which is a continuation of U.S. application Ser. No. 16/866,392, filed May 4, 2020, now U.S. Pat. No. 11,548,356, issued Jan. 10, 2023, which relates to and claims the benefit of U.S. Provisional Application No. 62/987,726, filed Mar. 10, 2020 and entitled "PROTECTIVE BARRIER FOR SAFETY GLAZING," the entire contents of which is expressly incorporated herein by reference.

STATEMENT RE: FEDERALLY SPONSORED RESEARCH/DEVELOPMENT

(1) Not Applicable

BACKGROUND

1. Technical Field

(2) The present disclosure relates generally to safety glazing and, more particularly, to a protective barrier applied to the exterior of a vehicle windshield.

2. Related Art

(3) Currently, vehicle windshields are being manufactured to include opto-electric devices such as cameras, rain sensors, proximity sensors, heads-up displays, defrosters, and antennas. This has increased the cost of replacing a cracked windshield by a factor of 10. In addition, costly calibration procedures must be performed after the installation of a new windshield, further increasing the costs associated with replacing the windshield.

(4) The American National Standards Institute (ANSI) Z26.1-1996 standard, entitled "Safety Glazing Materials for Glazing Motor Vehicles and Motor Vehicle Equipment Operating on Land Highways," is a standard that specifies the durability and safety requirements to qualify materials for vehicle glazing. Among the various tests mandated by the standard are transmission, humidity, heat, impact, fracturing, penetration, distortion, weathering, haze, and abrasion resistance.

Applicable standards such as the Z26.1-1996 standard currently specify sixteen categories of construction for safety glazing allowable in various locations on the vehicle. The most stringent category is for windshields because of the need for visual acuity, impact resistance, and abrasion resistance from pitting and wiper blades, as well as the need to contain glass fragments to prevent them from injuring passengers.

(5) The two basic groups of construction materials for windshields are glass and plastics. Per applicable standards, plastics are relegated to vehicles such as motorcycles, and the windshield is only allowed to be 15 inches higher than the seat so that the rider can look over it. This is because the plastics available today are so soft that they abrade easily, reducing visual acuity of the windshield after only a short service life. Considering their limited use, the abrasion test for plastics (e.g. test 5.17 specified in the ANSI Z26.1-1996 standard) only requires 100 cycles of Taber abrasion. This is 10 times less than the requirement for glass, which is 1,000 cycles of Taber abrasion. On the other hand, plastic could be a preferred material because it does not produce the sharp fragments that glass does with impacts and is half the weight. Although glass is hard and abrasion resistant, its low tensile strength makes it subject to pitting. In addition, on impact it produces dangerous sharp fragments that can injure the passengers. To mitigate this safety issue, glass windshields may be laminated with a soft plastic core to hold the fragments together and improve penetration resistance.

(6) There are no commercially available safety barrier films for windshields that meet applicable requirements for weathering for one year (e.g. test 5.16 specified in the ANSI Z26.1-1996 standard), abrasion of plastics (e.g. test 5.17 specified in the ANSI Z26.1-1996 standard), and abrasion of glass (e.g. test 5.18 specified in the ANSI Z26.1-1996 standard), as shown in the table below:

(7) TABLE-US-00001 ANSI Z26 FOR WINDSHIELDS Test Initial Haze after Taber Outdoor
Material # Haze Weathering Revolutions Exposure Plastic 5.17.3 N/A 15% 100 1 year Hard Plastic 5.17.4 1% 10% 500 1 year Glass/Plastic 5.17.3 N/A 4% 100 Plastic on Interior Glass 5.18 N/A 2% 1,000 1 year

(8) In today's commercial market, the existing polymer safety films for protecting glass windows are mounted to the interior of the building or vehicle. These commercial products would not last more than a few months if mounted to the exteriors, due to the embrittlement of the polyester substrate. This allows fracturing of the hard coating on the surface, and thus a failure of the film. The durability of interior mounted safety films to windows is specified in ANSI Z97.1-2015, entitled "American National Standard for Safety Glazing Materials Used in Buildings-Safety Performance Specifications and Methods of Test." An example of such an interior mounted safety film is a 3M Scotch-shield safety and security window film Ultra Series, having a thickness of 8

mil, a peel strength of 2,000 g/in, a transmission of greater than 88%, substantially no distortion at 45 degrees angle of incidence, and an abrasion resistance of 5% haze after 100 Taber cycles. While such a film may have a service life of 10 years, it is not manufactured for exterior usage and the abrasion resistance does not qualify for ANSI Z26 windshield usage (e.g. less than 2% haze after 1,000 Taber cycles).

(9) In the case of interior mounting of a safety film such as the above 3M film, the glass window itself may provide protection from portions of the UV and IR spectrum. The addition of UV inhibitors mixed into the mounting adhesive may be enough to provide a long service life (e.g. 10 years) of the film and hard coated surface facing the interior. However, the exterior glass surface remains subject to pitting due to the low tensile strength of the glass.

(10) The only commercially available safety film for exterior use on vehicles operating on land highways is made by Clear-Plex. According to Clear-Plex's published commercial specification and related U.S. Pat. Nos. 7,992,917 and 9,023,162, the Clear-Plex safety film includes a layer of 4 mil thick PET with a hard coat and pressure sensitive adhesive for mounting and has a peel strength of 1,800 g/in, a transmission of greater than 87%, substantially no distortion at 40 degrees angle of incidence, and an abrasion resistance of 0.5% haze after 100 cycles before weathering. Clear-Plex does not claim any testing performed to ANSI Z26 standards. The haze value may be acceptable for plastic, but the commercial specification does not include Taber testing after 1 year of weathering.

(11) There exist other products that may be mounted onto windshield exteriors of vehicles that do not operate on land highways, such as vehicles used in stockcar racing or military vehicles. One such product, by Racing Optics, Inc., is a 4-layer×4 mil safety film (hereinafter "RO 4×4") having a thickness of 18 mil (4 layers of 4 mil thick PET with a hard coat and pressure sensitive adhesive on each layer), a peel strength of 100 g/in for the upper layers and 400 g/in for the base layer, a transmission of greater than 88%, a Z26 haze test #5.17 result of less than 1.5% haze before weathering, a Z26 abrasion test #5.16 result of less than 5% haze after 100 Taber cycles, and a Z26 weatherability test #5.15 result of less than 4 months per layer (e.g. transmittance noticeably reduced after 3-4 months weathering, resulting in a haze of about 20-50% without Taber testing). Because the RO 4×4 product is designed for a short-term service life (which is renewed as each layer is peeled off during use), it does not have the weather durability or abrasion resistance required to meet the Z26 standard for a windshield of a vehicle operating on land highways.

BRIEF SUMMARY

(12) The present disclosure contemplates various systems and methods for overcoming the above drawbacks accompanying the related art. According to one or more aspects of the present disclosure, an external barrier may be added to both glass and plastic windshields to improve abrasion resistance, pitting, and impact cracking. The external barrier may increase the safety of passengers as well as reduce the number one insurance cost of the national fleet of vehicles, namely windshield damage, all while reducing the carbon footprint of replacing a glass windshield.

(13) One aspect of the embodiments of the present disclosure is a protective barrier affixable to a curved substrate. The protective barrier may comprise a stack of two or more lenses, each of the two or more lenses including a polyethylene terephthalate (PET) film, a hard coat on a first side of the PET film, and an adhesive layer on a second side of the PET film opposite the first side. The stack of two or more lenses may have a modulation transfer function that exhibits a contrast value greater than 75% for a spatial resolution of one line-pair per 0.0003 radians at 65 degrees angle of incidence.

(14) The modulation transfer function of the stack of two or more lenses may exhibit a contrast value greater than 70% for a spatial resolution of one line-pair per 0.0003 radians at 70 degrees angle of incidence. The modulation transfer function of the stack of two or more lenses may exhibit a contrast value greater than 85% for a spatial resolution of one line-pair per 0.0003 radians at 55 degrees angle of incidence. The modulation transfer function of the stack of two or more lenses may exhibit a contrast value greater than 90% for a spatial resolution of one line-pair per 0.0003

radians at 45 degrees angle of incidence.

(15) The PET film of each of the two or more lenses may have a modulation transfer function that exhibits a contrast value greater than 80% for a spatial resolution of one line-pair per 0.0003 radians at 65 degrees angle of incidence.

(16) Each of the two or more lenses may be 2-4 mil thick.

(17) The PET film of each of the two or more lenses may include UV stabilizers. The hard coat and the adhesive layer of each of the two or more lenses may include UV stabilizers.

(18) The PET film of each of the two or more lenses may have a machine direction shrinkage of 0.6%-1.8% and a transverse direction shrinkage of 0.3%-1.1% at 150° C.

(19) Another aspect of the embodiments of the present disclosure is a method. The method may comprise stacking two or more lenses, each of the two or more lenses including a polyethylene terephthalate (PET) film, a hard coat on a first side of the PET film, and an adhesive layer on a second side of the PET film opposite the first side. The stack of two or more lenses may have a modulation transfer function that exhibits a contrast value greater than 75% for a spatial resolution of one line-pair per 0.0003 radians at 65 degrees angle of incidence. The method may comprise placing the stack of two more lenses on a curved substrate with the adhesive of a first lens of the stack in contact with the curved substrate and applying heat and pressure to conform the stack of two or more lenses to the shape of the curved substrate.

(20) The applying of heat and pressure may be performed at least in part prior to the adhesive layer of each of the two or more lenses being fully cured. The applying of heat and pressure may be performed at least in part prior to the adhesive layer of each of the two or more lenses exceeding a peel strength of 25 grams per inch determined as a constant load per unit width needed for peeling.

(21) The method may comprise peeling off an outermost lens of the stack of two or more lenses after the applying of heat and pressure.

(22) The adhesive of the first lens of the stack of two or more lenses may be stronger than the adhesive of an outermost lens of the stack of two or more lenses.

(23) The modulation transfer function of the stack of two or more lenses may exhibit a contrast value greater than 70% for a spatial resolution of one line-pair per 0.0003 radians at 70 degrees angle of incidence.

(24) The PET film of each of the two or more lenses may have a modulation transfer function that exhibits a contrast value greater than 80% for a spatial resolution of one line-pair per 0.0003 radians at 65 degrees angle of incidence.

(25) Each of the two or more lenses may be 2-4 mil thick.

(26) The PET film of each of the two or more lenses may include UV stabilizers. The hard coat and the adhesive layer of each of the two or more lenses may include UV stabilizers.

(27) The PET film of each of the two or more lenses may have a machine direction shrinkage of 0.6%-1.8% and a transverse direction shrinkage of 0.3%-1.1% at 150° C.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and other features and advantages of the various embodiments disclosed herein will be better understood with respect to the following description and drawings, in which like numbers refer to like parts throughout, and in which:

(2) FIG. 1 is a cross-sectional view of a protective barrier according to an embodiment of the present disclosure;

(3) FIG. 2 is an image and graphical representation of visual distortion in a safety film;

(4) FIG. 3 is a graphical representation of modulation transfer function (MTF) data for five samples at different angles of incidence;

(5) FIG. 4 is a graphical representation of windshield damage velocity for different film thicknesses;

(6) FIG. 5 shows the protective barrier placed on a windshield of a car at the beginning of a process of applying heat and pressure to mold the protective barrier to the shape of the windshield;

(7) FIG. 6 shows the protective barrier on the windshield at the end of the process of applying heat and pressure;

(8) FIG. 7 shows the protective barrier after it has been trimmed to fit the windshield; and

(9) FIG. 8 is an example operational flow according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(10) The present disclosure encompasses various embodiments of a protective barrier affixable to a curved substrate and methods of manufacture, installation, and use thereof. The detailed description set forth below in connection with the appended drawings is intended as a description of several currently contemplated embodiments and is not intended to represent the only form in which the disclosed invention may be developed or utilized. The description sets forth the functions and features in connection with the illustrated embodiments. It is to be understood, however, that the same or equivalent functions may be accomplished by different embodiments that are also intended to be encompassed within the scope of the present disclosure. It is further understood that relational terms such as first and second and the like are used solely to distinguish one from another entity without necessarily requiring or implying any actual such relationship in order between such entities.

(11) FIG. 1 is a cross-sectional view of a protective barrier **100** according to an embodiment of the present disclosure. The protective barrier **100** may be affixed to a curved substrate **10** such as a windshield of an automobile and may comprise a stack of two or more lenses **110** such as the lenses **110a**, **110b**, **110c** shown in FIG. 1. Each of the lenses **110** may include a polyethylene terephthalate (PET) film **112**, a hard coat **114** on a first side of the PET film **112**, and an adhesive layer **116** on a second side of the PET film **112** opposite the first side for bonding the lenses **110** together and to the curved substrate **10**. The stack of lenses **110** may have a modulation transfer function at 65 degrees angle of incidence that exhibits a contrast value greater than 75% for a spatial resolution of one line-pair per 0.0003 radians, the approximate resolution of the human eye. By controlling the modulation transfer function of the stack of lenses **110** in this way, a protective barrier **100** may be produced that is substantially distortion-free (e.g. less than 0.00045 radians displacement) when viewed at an angle of incidence typical of automobile windshields (e.g. 60-70 degrees), even while the total thickness of the protective barrier **100** is sufficient to resist impact damage at automobile speeds. In this way, the protective barrier **100** can prevent cracking and pitting of the underlying windshield **10** while meeting the durability requirements for windshield use on highways.

(12) Distortion is a visual acuity error caused by a displaced object in the far field (e.g. 40-1,000 feet). Safety glazing can have localized zones which cause object displacements perceived as distortion, where an object may appear to jump from one position to another when viewed from a slightly different position or angle. Conventionally, distortion is only qualitatively determined, for example, by test 5.15 specified in the ANSI Z26.1-1996 standard. This test uses a shadow graph in a long tunnel with a collimated light source and white screen. The technician places a specimen in the light path at normal angle of incidence, 15 inches from the screen. The technician then looks for dark and light artifacts caused by distortion. The test has no quantitative criteria and does not address distortion at the high angle of incidence used in modern car windshields (e.g. 60-70 degrees).

(13) Ideally, the distortion should be minimized to the resolution of the human eye with 20/20 vision, which is about one line-pair per 0.0003 radians. If an object is displaced by 0.0006 radians, for example, then the eye will perceive the location change as distortion. Effectively, then, distortion must be reduced such that any object displacement is below the resolution capability of

the human eye in order to be considered distortion-free. Meanwhile, when safety glazing is viewed at high angles of incidence (e.g. 60-70 degrees), the optical thickness increases as a function of the cosine of the angle according to Snell's law. This may cause any distortion effects to be amplified, especially for safety film thicker than around 4 mil as may be needed to resist impact damage at automobile speeds.

(14) FIG. 2 is an image and graphical representation of visual distortion in a safety film **210**. In order to quantitatively measure distortion, it is contemplated that the modulation transfer function of a test material such as the safety film **210** may be evaluated at a fixed spatial frequency of 0.0003 radians, corresponding to the resolution of the human eye with 20/20 vision. To this end, as shown in the upper part of FIG. 2, an image of a test pattern **220** such as a checkerboard pattern or target line-pairs spaced by 0.0003 radians may be captured through the film **210** at a desired angle of incidence (e.g. 65 degrees in FIG. 2). The lower part of FIG. 2 shows the corresponding modulation transfer function data representing the contrast as a function of horizontal position for a given cross-section of the data representing a single horizontal slice of the image. As can be seen, the modulation transfer function data exhibits reduced contrast, corresponding to distorted line-pairs, in the area of the test pattern **220** that is viewed through the film **210**. In some places, the modulation transfer function data exhibits such low contrast as to amount to complete loss of the image.

(15) A test setup such as that of FIG. 2 may be used to evaluate materials and process parameters for manufacturing the protective barrier **100** described herein. In particular, by using such a test setup and/or test results derivable therefrom, appropriate materials and process parameters may be selected and/or adjusted to control the modulation transfer function of the stack of lenses **110** at one or more desired angles of incidence. In this regard, it is contemplated that the modulation transfer function of the stack of lenses **110** may be controlled according to the methodology described in commonly owned U.S. Provisional Application No. 62/942,943, filed Dec. 3, 2019 and entitled "METHOD AND APPARATUS FOR REDUCING NON-NORMAL INCIDENCE DISTORTION IN GLAZING FILMS," the entire contents of which is expressly incorporated herein by reference.

(16) For example, at any or all stages of producing the stack of lenses **110** (e.g. during the formation of the PET film **112** by melting a resin, extruding the melted resin through a die to produce a film, and cooling the film, during the applying of the hard coat **114**, during the applying of the adhesive layer **116**, etc.), one or more images of the test pattern **220** may be captured through the lens **110** or stack of lenses **110** being produced. The image(s) may be captured, for example, by aiming an image capturing device through a roll-to-roll processing web containing the lens **110** or stack of lenses **110** at one or more desired angles of incidence. On the basis of such image(s), a computer may calculate the MTF data and produce an output used for adjusting process parameters that are found to effect the modulation transfer function of the lens **110** or stack of lenses **110**, such as a temperature setting of a heater used in melting the resin (e.g. absolute temperature or relative temperatures of a gradient or profile of a plurality of heated regions of an extruder assembly), a rotation speed of an extrusion screw (which may determine melting time as well as degree of mixing of the resin), a rotation speed of one or more rollers (which may determine cooling time and/or a degree of force acting on the polymer film during cooling), a flow speed, deposition speed, or other application speed of the hard coat **114** or adhesive layer **116**, and/or a speed at which the lenses **110** are stacked. It is contemplated, for example, that the PET film **112** may in some cases be prefabricated and selected for its known MTF data, whereas the modulation transfer function of the stack of lenses **110** may be actively controlled during the application of the hard coats **114** and/or adhesive layers **116** and the stacking of the lenses **110**. In other cases, the PET film **112** may also be manufactured while actively controlling the modulation transfer function thereof. The output of the computer may include, for example, a feedback signal for automatically adjusting the relevant process parameters without user input in either a continuous or batch-to-batch process. As another example, the output may include a visual representation of the data to be interpreted by an operator

who will make the necessary adjustments manually.

(17) FIG. 3 is a graphical representation of modulation transfer function data for five samples at different angles of incidence. To produce the example data of FIG. 3, modulation transfer function data as described in FIG. 2 may be taken at angles of incidence from normal (zero degrees) to 70 degrees in increments of 10 degrees. It is contemplated that the data may be normalized to a modulation transfer function value representing a windshield without any protective barrier. Because many car windshields are installed with a slant of 65 degrees, additional data may be captured at 65 degrees, or likewise any other angle of particular interest. As represented in FIG. 3 by the solid line with triangular datapoints, the sample labeled T-11 3×3 may serve as the stack of lenses **110** of the protective barrier **100** described herein, having a modulation transfer function that exhibits a contrast value greater than 75% for a spatial resolution of one line-pair per 0.0003 radians at 65 degrees angle of incidence. As noted above, 65 degrees is a typical windshield slant. However, a driver must also observe objects above or below eye level, such that it may also be advantageous to minimize distortion at larger or smaller angles of incidence. To this end, as shown in the example of the sample labeled T-11 3×3, the modulation transfer function of the stack of lenses **110** may further exhibit, for the same spatial resolution of one line-pair per 0.0003 radians, a contrast value greater than 70% at 70 degrees angle of incidence, a contrast value greater than 85% at 55 degrees angle of incidence, and/or a contrast value greater than 90% at 45 degrees angle of incidence. A protective barrier **100** whose modulation transfer function is controlled in this way may be applied to a typical automobile windshield without distorting the positions of objects viewed by the driver.

(18) An example of a prefabricated PET film **112** that may be selected for its known MTF data for use in the stack of lenses **110** is a PET film **112** having a modulation transfer function that itself exhibits a contrast value greater than 80% for a spatial resolution of one line-pair per 0.0003 radians at 65 degrees angle of incidence. One such material is a film sold under the name MELINEX® 454 by DuPont Teijin Films, which is represented in FIG. 3 by the solid line with circular datapoints. The dashed line with diamond datapoints represents a sample of a lens **110** made using this film as the PET film **112** thereof, with the hard coat **114** and the adhesive layer **116** having been applied under MTF control as described above. That is, during the processes of applying the hard coat **114** and the adhesive layer **116**, one or more process parameters were selected or adjusted (either continuously or batch-to-batch) to control the modulation transfer function at one or more angles of incidence, for example, to maintain a contrast value greater than 75% for a spatial resolution of one line-pair per 0.0003 radians at 65 degrees angle of incidence. As can be seen, the MTF data of this sample, which represents only a single lens **110** in accordance with the disclosed subject matter, is substantially similar to that of the sample labeled T-11 3×3, which represents the entire protective barrier **100** including a stack of such lenses **110**, despite the increased thickness of the stack. This may be achieved by stacking the lenses **110** under MTF control as described above.

(19) In contrast, the sample labeled T-8 3×3, represented in FIG. 3 by the dash-dot line with square datapoints, was produced without MTF control. Despite being a similarly constructed 3-layer stack, the optical properties are markedly inferior at the high angles of incidence typically used in car windshields. For example, as shown in FIG. 3, the contrast value at 65 degrees angle of incidence is under 60%. Such a product can only be used in applications having an angle of incidence less than 60 degrees, where the contrast value remains above 75%. The sample labeled RO 4×4, represented by the solid line with circular datapoints and corresponding to the RO 4×4 product described above, is a 4-layer stack that was similarly produced without MTF control. This product exhibits even worse MTF data at relevant angles of incidence and can realistically only be used in applications where the angle of incidence is less than 50 degrees, after which the contrast value falls below 75%. It has been found that distortion may become noticeable below a contrast value of 75%.

(20) FIG. 4 is a graphical representation of windshield damage velocity for different film thicknesses. The example data of FIG. 4 is based on the results of a glass fracture study conducted by O'Gara-Hess Armor Company for the U.S. military. Increasingly thick layers of PET were mounted on ballistic glass, and a three-quarters inch steel ball was launched at the glass at different velocities. The data shows the minimum velocity that resulted in glass cracking for each PET film thickness. In light of the data shown in FIG. 4, it is contemplated that the protective barrier **100** described herein should be 8 mil thick or thicker in order to protect a glass windshield at common driving velocities of 45-65 miles per hour, for example, 8 mil to 16 mil thick and preferably 10 mil to 16 mil thick. For example, the protective barrier **100** may include 2-4 lenses **110** (e.g. 3 lenses **110a**, **110b**, and **110c** as shown in FIG. 1), where each of the lenses **110** is 4 mil thick.

(21) In general, the increased thickness needed to protect a glass windshield poses several challenges to the production of the protective barrier **100**. As described above, for example, the increased thickness may amplify distortion at high angles of incidence (e.g. 60-70 degrees). This challenge may be overcome by controlling the modulation transfer function of the protective barrier **100** as described above, for example, by producing a stack of lenses **110** having a modulation transfer function that exhibits a contrast value greater than 75% for a spatial resolution of one line-pair per 0.0003 radians at 65 degrees angle of incidence. Additional challenges posed by the thickness of the protective barrier **100** include making a product that can be successfully molded to a curved substrate **10** (e.g. an automobile windshield), achieving a high degree of weatherability and abrasion resistance and reduced haze, and maintaining a reasonably long service life. Each of these challenges may be overcome by the disclosed protective barrier **100** as described in more detail below.

(22) FIG. 5 shows the protective barrier **100** placed on a windshield **10** of a car **20** at the beginning of a process of applying heat and pressure to mold the protective barrier **100** to the shape of the windshield **10** (the windshield **10** serving as the substrate **10** shown in FIG. 1). The protective barrier **100** may be adhered to the windshield **10** by placing the adhesive layer **116** of a first (bottommost) lens **110a** of the stack in contact with the windshield **10** (see FIG. 1). The adhesive layer **116** of the first lens **110a** may be a dry mount adhesive as disclosed, for example, in U.S. Pat. No. 9,295,297 to Wilson, issued Mar. 29, 2016 and entitled "Adhesive Mountable Stack of Removable Layers," the entire contents of which is expressly incorporated herein by reference. Alternatively, a wet mount adhesive may be used as disclosed, for example, in U.S. Pat. No. 9,128,545 to Wilson, issued Sep. 8, 2015 and entitled "Touch Screen Shield," the entire contents of which is expressly incorporated herein by reference. The adhesive may be an acrylic adhesive such as an acrylic pressure sensitive adhesive (PSA).

(23) The windshields of most cars exhibit a compound curvature, such that the protective barrier **100** will not conform to the windshield **10** without shrinking it in the upper and lower corners. Since the protective barrier **100** may be flat (e.g. having been manufactured in a roll-to-roll process), the stack of two or more lenses **110** may not initially conform to the curved shape of the windshield **10**, resulting in regions of greater or less adhesion and pockets/bubbles of air between the stack of lenses **110** and the windshield. Therefore, in order to conform the stack of lenses **110** to the shape of the windshield **10**, heat and pressure may be applied using a heater **30** such as a hot air source (e.g. a heat gun or blow dryer) or an infrared heater. At the same time, pressure may be applied to the stack of lenses **110** using a card or squeegee. In some cases, the protective barrier **100** may be applied using a sacrificial layer serving as a female mold cavity to sandwich the stack of lenses **110** between the sacrificial layer and the windshield **10** as described in commonly owned U.S. application Ser. No. 16/778,928, filed Jan. 31, 2020 and entitled "THERMOFORM WINDSHIELD STACK WITH INTEGRATED FORMABLE MOLD," the entire contents of which is expressly incorporated herein by reference.

(24) As the installer heats and presses down on the stack of lenses **110**, the stack of lenses **110** may shrink and stretch to take on the contour of the curved substrate **10** (the windshield). In the case of

commercial films having a thickness of only 2 mil, the necessary shrinking may be easy to achieve. On the other hand, a monolithic film of 8 mil or thicker will crease before the film conforms to the windshield, making it unusable. In light of this challenge, the protective barrier **100** described herein uses multiple thin lenses **110** (e.g. 2-4 mil thick each) that individually shrink well. The adhesive layers **116** between the lenses **110** of the stack, which may be the same acrylic adhesive for example, may be only partially cured to produce an extremely low peel strength (e.g. 15-25 g/in) and high elasticity. As such, each individual lens **110** of the stack may “float” in relationship to each other, allowing shrinking to occur without creasing any of the lenses **110**. Once the protective barrier **100** is installed and exposed to sunlight, for example, the adhesive layers **116** will cure and increase the peel and bonding strength (e.g. by a factor of 3 to 5), promoting a long service life. The peel strength after initial weathering may be 100-150 g/in, for example.

(25) FIG. **6** shows the protective barrier **100** on the windshield **10** at the end of the process of applying heat and pressure. At this stage, the desired shrinkage has occurred and the stack of lenses **110** of the protective barrier **100** is molded to the curved shape of the windshield **10** without air pockets/bubbles. The technical specifications of PET films include two axes of shrinking with different values, designated as the machine direction (“MD”) and the transverse direction (“TD”). The machine direction refers to the direction of the roll stock in a roll-to-roll process used to produce the PET film, while the transverse direction refers to the direction across the roll direction. It is contemplated that the PET films **112** of the stack of the protective barrier **100** may have a machine direction shrinkage of 0.6%-1.8% (preferably 0.8%-1.0%) and a transverse direction shrinkage of 0.3%-1.1% (preferably 0.5%-0.6%) at 150° C. A PET films that has a machine direction shrinkage below 0.6% or a transverse direction shrinkage below 0.3% will not have enough shrink to conform to a windshield. On the other hand, if the shrinkage is too high, e.g. greater than 1.8% in the machine direction or greater than 1.1% in the transverse direction, it will be too difficult for the installer to control the shrinkage in a hand-operated procedure (e.g. using the heater **30** as described above).

(26) FIG. **7** shows the protective barrier **100** including the stack of lenses **110** after the stack of transparent lenses **110** has been trimmed to fit the windshield **10**. The stack of lenses **110** may be trimmed using a knife such as a utility knife or box cutter with a stainless-steel blade (a carbon blade may damage the windshield **10**). The resulting trimmed stack of lenses **110** may effectively be invisible as it matches the shape of the windshield **10** beneath (though it may alter the coloring of the windshield as in the case of window tinting).

(27) In addition to improving moldability as described above, the use of multiple thin lenses **110** (e.g. 2-4 mil thick) rather than a single monolithic film may allow for a sufficiently reduced haze to be usable on automobile windshields. In general, haze in a PET film has two components: scattering of incident light at the surface and dispersion of incident light in the bulk material. The latter bulk component increases with the thickness of the PET film, for example, as shown in the table below:

(28) TABLE-US-00002

Thickness	Haze %
2 mil	0.4
4 mil	0.6
7 mil	0.8
10 mil	1.2

(29) However, the effect is not additive when multiple PET films are stacked, with three layers only adding about 0.1-0.2% haze in total. Meanwhile, the surface component of haze is mitigated by the addition of hard coat or adhesive. By structuring the protective barrier **100** as a stack of lenses **110** containing relatively thin PET films **112** rather than a single large PET film, a reduced haze can be achieved even while the protective barrier **100** may be thick enough to resist impact at automobile speeds (e.g. 8 mil or thicker) as described above. In particular, the protective barrier **100** described herein, comprising a stack of two or more lenses **110** each of which includes a PET film **112** with a hard coat **114** and an adhesive layer **116**, may achieve an initial (pre-weathering) haze of below 1% (preferably below 0.6%), making it suitable for use on automobile windshields.

(30) Weathering may be defined according to a standard such as the ANSI Z26.1-1996 standard at exposure of one year at around 300 MJ/m.² of ultraviolet radiation (e.g. 301 MJ/m.² or 306

MJ/m.sup.2 per applicable standards, or 280 MJ/m.sup.2 extrapolated from 70 MJ/m.sup.2 per three month period). In order to simulate exposure for one year in an outdoor Arizona climate (Arizona being selected as a weathering benchmark for its high temperatures and high-intensity sunlight), a natural sunlight concentrator may be used such as one that complies with the American Society for Testing and Materials (ASTM) G90 standard, entitled "Standard Practice for Performing Accelerated Outdoor Weathering of Materials Using Concentrated Natural Sunlight." The haze and abrasion resistance of the protective barrier **100** may be measured before and at the end of the exposure cycle.

(31) In the comparative example of the RO 4×4 product described above, a UV stabilizer such as a UV absorbing compound is mixed into the hard coat and the adhesive of each of the four layers. After only six months of Arizona exposure, the outermost layer becomes unusable because of loss of transmission, increased haze, and loss of hardness. Because there is so much UV inhibitor in the hard coat, the hard coat has reduced hardness and fractures, allowing the underlying PET core of the outermost layer to become yellow and brittle. The resulting haze may exceed 20%.

(32) In contrast to the RO 4×4 product, the protective barrier **100** described herein may be produced with the PET film **112** of each of the two or more lenses **110** including UV stabilizers such as hydroxyphenyl-benzotriazole or hydroxyphenyl-triazine UV absorbers. The hard coat **114** and/or the adhesive layer **116** of each lens **110** may also include UV stabilizers. Because the UV stabilizers are mixed into the PET film **112**, a reduced amount of UV stabilizers can be used in the hard coat **114** and adhesive layer **116**, allowing the hard coat **114** to maintain its hardness without sacrificing UV stability. Spreading the UV stabilizers across all components allows for a highly weatherable assembly, such that the protective barrier **100** may exceed one year of ANSI G90 exposure in Arizona sun and may look very good with low haze and little if any yellowing after weathering. The protective barrier **100** may, for example, have an abrasion resistance at 1,000 Taber cycles of less than 1% haze before weathering and less than 4% (preferably less than 2%) haze after weathering.

(33) Although the protective barrier **100** may have sufficient weatherability, eventually the outermost lens **110** (e.g. lens **110c** in the 3-layer example of FIG. 1) may become damaged. When the outermost lens **110** becomes unacceptably degraded over time during the life of the vehicle windshield or other window (e.g. due to chips, oxidation, etc.), the outermost lens **110** may simply be peeled off and removed, revealing a fresh lens **110** beneath. To this end, the adhesive layer **116** of the innermost lens **110a** (see FIG. 1) may be stronger than the adhesive layers **116** used for the other lenses **110** (and in some cases the adhesive layers **116** may have further decreasing strength with each additional lens **110**). In this way, the innermost lens **110a** may remain adhered to the windshield or other curved substrate **10** while another lens **110** is peeled off. It is contemplated, for example, that the innermost lens **110a** may be intended to remain on the curved substrate **10** for the life of the protective barrier **100**, with additional lenses **110** being removable as needed. Along the same lines, each such additional lens **110** beyond the first **110a** may be provided with a tab or other means for easy peel-away during the life of the protective barrier **100**. By allowing for peeling away of the outermost lens **110** of the stack of lenses **110** in this way, the service life of the protective barrier **100** may be extended.

(34) FIG. 8 is an example operational flow according to an embodiment of the present disclosure. The operational flow of FIG. 8 may serve as an example method of manufacturing, installing, and using the protective barrier **100** including the stack of lenses **110** shown in FIG. 1. The operational flow may begin with providing a PET film **112** to be used as the core of each of two or more lenses **110** (step **810**). As explained above, the PET film **112** of each of the lenses **110** may be selected for particular MTF data, such as a contrast value greater than 80% for a spatial resolution of one line-pair per 0.0003 radians at 65 degrees angle of incidence and may be, for example, a film sold under the name MELINEX® 454 by DuPont Teijin Films. Alternatively, the PET film **112** of each of the lenses **110** may be fabricated while actively monitoring the MTF data in a continuous or batch-to-

batch process as described above. In this regard, providing the PET film **112** may include, for example, melting a resin, extruding the melted resin through a die to produce a polymer film, and cooling the polymer film. A hard coat **114** may be deposited on a first side of the PET film **112** (step **820**), which is preferably wet deposited but may be applied according to any appropriate methods including spin coating, dip coating, or vacuum deposition. Before or after the hard coat **114** is applied, the PET film **112** may be coated on the opposite side with an adhesive **116** (step **830**). These three elements, the PET film **112**, hard coat **114**, and adhesive **116** may constitute one of the lenses **110** described herein, which may be stacked to produce the protective barrier **100** (step **840**).

(35) During any or all of steps **810-840**, the operational flow may comprise controlling the MTF of the stack of lenses **110** (step **850**). The MTF of the stack of lenses **110** may be controlled, for example, so as to exhibit a contrast value greater than 75% for a spatial resolution of one line-pair per 0.0003 radians at 65 degrees angle of incidence. As explained above, such control may be achieved by selecting an appropriate pre-fabricated PET film **112** in step **810**. Alternatively, or additionally, the control of the MTF may be achieved by actively monitoring and adjusting process parameters (e.g. roller speed of a roll-to-roll process, etc.) while fabricating a PET film **112** in step **810**, depositing the hard coat **114** in step **820**, applying the adhesive layer **116** in step **830**, and/or stacking the two or more lenses **110** in step **840**. It is contemplated that such active monitoring and adjusting of process parameters may include a continuous process including a feedback loop of monitored MTF data and/or a batch-to-batch process with MTF measurements manually or automatically fed back from a preceding batch.

(36) Once the protective barrier **100** comprising the stack of lenses **110** has been assembled, the operational flow may continue with installing the protective barrier **100** on a curved substrate **10** such as the windshield of the car **20** shown in FIGS. 5-7. As explained above, the installing may be done while the adhesive layers **116** are only partially cured in order to allow the lenses **110** to “float” on the adhesive and mold to the shape of the substrate **10** individually, rather than as a unitary structure, to avoid creasing. Referring to the operational flow of FIG. 8, the protective barrier **100**, including the stack of lenses **110**, may be placed on the windshield or other curved substrate **10** (step **860**), with the adhesive layer **116** of the lowermost lens **110a** (see FIG. 1) in contact with the curved substrate **10**. For easier installation, the protective barrier **100** may be rough cut (e.g. using an electric film cutter) so as not to extend too far outside the windshield **10**. The operational flow may continue with applying heat and pressure to conform the stack of two or more lenses **110** to the shape of the curved substrate **10** (step **870**) as described in relation to FIGS. 5 and 6. In particular, the applying of heat and pressure may be performed at least in part prior to the adhesive layer **116** of each of the two or more lenses **110** being fully cured, for example, prior to the adhesive layer **116** exceeding a peel strength of 25 grams per inch determined as a constant load per unit width needed for peeling. The protective barrier **100** may be completely conformed to the shape of the curved substrate **10** prior to the adhesive layers **116** being fully cured.

(37) After allowing the protective barrier **100** to cool down, the installation may conclude with performing a final trim as described in relation to FIG. 7. The protective barrier **100** including the stack of lenses **110** is now uniformly formed and affixed to the windshield surface. By having the protective barrier **100** installed in this way, rock strike cracking and abrasion damage to the windshield **10** can be reduced while still complying with applicable standards for windshield transmission, abrasion resistance, haze, and distortion for a vehicle operating on land highways.

(38) As explained above, it is contemplated that a protective barrier **100** having more than one lens **110** may allow for the outermost lens **110** to be peeled off and removed to reveal the unused surface of the lens **110** beneath. In this respect, the operational flow of FIG. 8 may continue during the life of the protective barrier **100** that has been installed on a vehicle **20**. When the outermost lens **110** becomes unacceptably degraded over time (e.g. after six months, after a year, after scratching from wiper blades begins to occur, etc.), it may be peeled off to reveal the next lens **110** underneath (step

880). The timing of peeling off the outermost lens **110** may depend on the particular climate where the protective barrier **100** is used, with some climates entailing more exposure to sun and others requiring more frequent use of wiper blades, for example.

(39) The above description is given by way of example, and not limitation. Given the above disclosure, one skilled in the art could devise variations that are within the scope and spirit of the invention disclosed herein. Further, the various features of the embodiments disclosed herein can be used alone, or in varying combinations with each other and are not intended to be limited to the specific combination described herein. Thus, the scope of the claims is not to be limited by the illustrated embodiments.

Claims

1. A protective barrier affixable to a curved substrate, the protective barrier comprising a stack of two or more lenses, each of the two or more lenses including a polyethylene terephthalate (PET) film and a hard coat on a first side of the PET film, the stack of two or more lenses having a modulation transfer function that exhibits a contrast value greater than 90% for a spatial resolution of one line-pair per 0.0003 radians at 45 degrees angle of incidence.
2. The protective barrier of claim 1, wherein each of the two or more lenses further includes an adhesive layer on a second side of the PET film opposite the first side.
3. The protective barrier of claim 2, wherein the adhesive layer of each of the two or more lenses includes UV stabilizers.
4. The protective barrier of claim 1, wherein the modulation transfer function of the stack of two or more lenses exhibits a contrast value greater than 80% for a spatial resolution of one line-pair per 0.0003 radians at 60 degrees angle of incidence.
5. The protective barrier of claim 1, wherein the modulation transfer function of the stack of two or more lenses exhibits a contrast value greater than 85% for a spatial resolution of one line-pair per 0.0003 radians at 55 degrees angle of incidence.
6. The protective barrier of claim 1, wherein the PET film of each of the two or more lenses has a modulation transfer function that exhibits a contrast value greater than 70% for a spatial resolution of one line-pair per 0.0003 radians at 70 degrees angle of incidence.
7. The protective barrier of claim 1, wherein each of the two or more lenses is 2-4 mil thick.
8. The protective barrier of claim 1, wherein the PET film of each of the two or more lenses includes UV stabilizers.
9. The protective barrier of claim 1, wherein the hard coat includes UV stabilizers.
10. The protective barrier of claim 1, wherein the PET film of each of the two or more lenses has a machine direction shrinkage of 0.6%-1.8% and a transverse direction shrinkage of 0.3%-1.1% at 150° C.
11. A method comprising: stacking two or more lenses, each of the two or more lenses including a polyethylene terephthalate (PET) film and a hard coat on a first side of the PET film, the stack of two or more lenses having a modulation transfer function that exhibits a contrast value greater than 90% for a spatial resolution of one line-pair per 0.0003 radians at 45 degrees angle of incidence; placing the stack of two more lenses on a curved substrate to contact with the curved substrate; and applying heat and pressure to conform the stack of two or more lenses to the shape of the curved substrate.
12. The method of claim 11, wherein each of the two or more lenses further includes an adhesive layer on a second side of the PET film opposite the first side.
13. The method of claim 12, wherein the adhesive layer of each of the two or more lenses includes UV stabilizers.
14. The method of claim 12, wherein said applying heat and pressure is performed at least in part prior to the adhesive layer of each of the two or more lenses being fully cured.

15. The method of claim 11, further comprising peeling off an outermost lens of the stack of two or more lenses after said applying heat and pressure.
 16. The method of claim 11, wherein the PET film of each of the two or more lenses has a modulation transfer function that exhibits a contrast value greater than 80% for a spatial resolution of one line-pair per 0.0003 radians at 60 degrees angle of incidence.
 17. The method of claim 11, wherein each of the two or more lenses is 2-4 mil thick.
 18. The method of claim 11, wherein the PET film of each of the two or more lenses includes UV stabilizers.
 19. The method of claim 11, wherein the hard coat of each of the two or more lenses includes UV stabilizers.
 20. The method of claim 11, wherein the PET film of each of the two or more lenses has a machine direction shrinkage of 0.6%-1.8% and a transverse direction shrinkage of 0.3%-1.1% at 150° C.
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